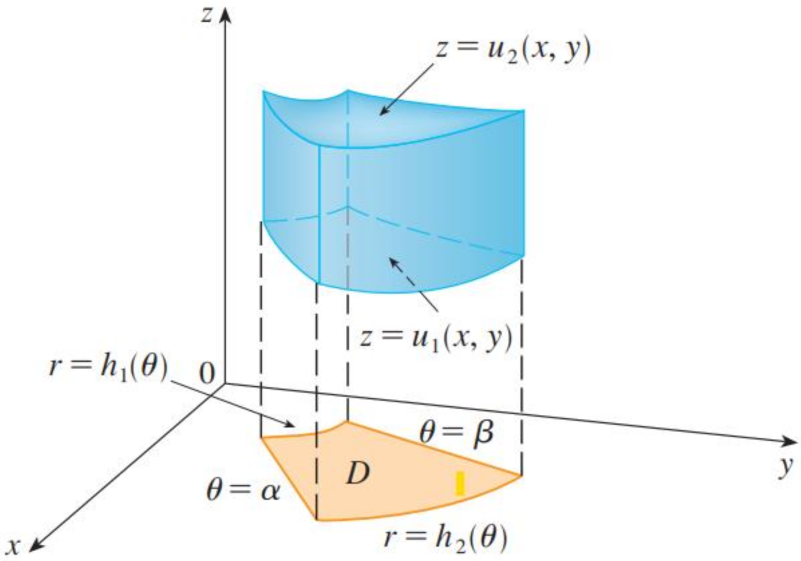
Triple integrals

$$\iiint_{\Omega} f(x,y,z)\,dV$$

If

$$\Omega = \{(x,y) \in D: u_1(x,y) \le z \le u_2(x,y)\}$$

$$\iiint_{\Omega} f(x,y,z)\,dV = \iint_{D}\left(\int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)dz\right)dA$$



Polar coordinates:

$$x = r\cos\theta \,, y = r\sin\theta \text{ and } z = z$$

With $\alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)$.

$$\iiint_{\Omega} f(x,y,z)\,dV = \int_{\alpha}^{\beta}\int_{h_1(\theta)}^{h_2(\theta)}\int_{u_1(r,\theta)}^{u_2(r,\theta)} f(r\cos\theta, r\sin\theta, z) r\,dz\,dr\,d\theta$$

a) $\Omega:\begin{cases} z = \sqrt{x^2+y^2} \\ z = 2 - x^2 - y^2 \end{cases}$

b) $\Omega:\begin{cases} z \le 4 - x^2 - y^2 \\ z \ge 0 \\ x^2 + y^2 \le 1 \end{cases}$

The volume of Ω

$$V(\Omega) = \iiint_{\Omega} 1dV$$

$$\Omega = \begin{cases} \sqrt{x^2+y^2} \le z \le 2 - x^2 - y^2 \\ x^2 + y^2 \le 1 \end{cases}$$

Change coordinates

$$\begin{cases} x = r\cos a \\ y = r\sin a \\ z = z \end{cases}, \qquad with\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1$$

$$\Omega = \begin{cases} r \le z \le 2 - r^2 \\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1 \end{cases}$$

According to, the volume is

$$\int_0^1 \int_0^{2\pi} \int_r^{2-r^2} 1 r dz da dr = ?$$

a) $\Omega : \begin{cases} z = \sqrt{x^2+y^2} \\ z = 2 - x^2 - y^2 \end{cases}$

b) $\Omega : \begin{cases} z \le 4 - x^2 - y^2 \\ z \ge 0 \\ x^2 + y^2 \le 1 \end{cases}$

$$\Omega = \begin{cases} 0 \le z \le 4 - x^2 - y^2 \\ x^2 + y^2 \le 1 \end{cases}$$

Change coordinates

$$\begin{cases} x = r\cos a \\ y = r\sin a \\ z = z \end{cases}, \qquad with\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1$$

$$\Omega = \begin{cases} 0 \le z \le 4 - r^2 \\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1 \end{cases}$$

Then

$$V(\Omega) = \int_0^{2\pi} \int_0^1 \int_0^{4-r^2} 1 r dz dr da = ?$$

c) $\Omega : \begin{cases} x^2 + y^2 \le 1 \\ z = 6 - \sqrt{x^2+y^2} \\ z = x^2 + y^2 \end{cases}$

d) $\Omega : \begin{cases} 1 \le x^2 + y^2 \le 4 \\ z \le 6 - \sqrt{x^2+y^2} \\ z \ge x^2 + y^2 \end{cases}$

$$\Omega = \begin{cases} x^2 + y^2 \le z \le 6 - \sqrt{x^2+y^2} \\ x^2 + y^2 \le 1 \end{cases}$$

Change coordinates

$$\begin{cases} x = r\cos a \\ y = r\sin a, \\ z = z \end{cases} \quad with\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1$$

$$\Omega = \begin{cases} r^2 \le z \le 6 - r \\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1 \end{cases}$$

Then

$$V(\Omega) = \int_0^{2\pi} \int_0^1 \int_r^{6-r} 1 r dz dr da =$$

$$\Omega = \begin{cases} x^2 + y^2 \le z \le 6 - \sqrt{x^2 + y^2} \\ 1 \le x^2 + y^2 \le 4 \end{cases}$$

Change coordinates

$$\begin{cases} x = r\cos a \\ y = r\sin a, \\ z = z \end{cases} \quad with\ 0 \le a \le 2\pi, \qquad 1 \le r \le 2$$

$$\Omega = \begin{cases} r^2 \le z \le 6 - r \\ 0 \le a \le 2\pi, \qquad 1 \le r \le 2 \end{cases}$$

Then

$$V(\Omega) = \int_0^{2\pi} \int_1^2 \int_r^{6-r} 1 r dz dr da =$$

e) $\Omega: \begin{cases} x^2 + y^2 + z^2 \le 4 \\ x \ge y^2 + z^2 \\ y^2 + z^2 \le 1 \end{cases}$

f) $\Omega: \begin{cases} x^2 + y^2 + z^2 \le 2x \\ x \ge 1 + \sqrt{y^2 + z^2} \end{cases}$

$$\Omega = \begin{cases} z^2 + y^2 \le x \le \sqrt{4 - y^2 - z^2} \\ z^2 + y^2 \le 1 \end{cases}$$

Change coordinates

$$\begin{cases} y = r\cos a \\ z = r\sin a, \\ x = x \end{cases} \quad with\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1$$

$$\Omega = \begin{cases} r^2 \le x \le \sqrt{4 - r^2} \\ 0 \le a \le 2\pi, \qquad 0 \le r \le 1 \end{cases}$$

Then $V(\Omega) = \int_0^{2\pi} \int_0^1 \int_{r^2}^{\sqrt{4-r^2}} 1rdzdrda = ?$

29–30 Evaluate the integral by changing to cylindrical coordinates.

29. $\displaystyle\int_{-2}^{2} \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_{\sqrt{x^2+y^2}}^{2} xz\, dz\, dx\, dy$

30. $\displaystyle\int_{-3}^{3} \int_{0}^{\sqrt{9-x^2}} \int_{0}^{9-x^2-y^2} \sqrt{x^2+y^2}\, dz\, dy\, dx$

$$\Omega = \begin{cases} \sqrt{x^2+y^2} \le z \le 2 \\ -\sqrt{4-y^2} \le x \le \sqrt{4-y^2} \\ -2 \le y \le 2 \end{cases} = \begin{cases} \sqrt{x^2+y^2} \le z \le 2 \\ x^2+y^2 \le 4 \end{cases}$$

Change coordinates

$$\begin{cases} x = rcosa \\ y = rsin\, a\,, \\ z = z \end{cases} \qquad with\ 0 \le a \le 2\pi, \qquad 0 \le r \le 2$$

$$\Omega = \begin{cases} r \le z \le 2 \\ 0 \le a \le 2\pi, \qquad 0 \le r \le 2 \end{cases}$$

Then

$$\int_0^2 \int_0^{2\pi} \int_r^2 (rcosa)zrdzdadr$$